

KENYA JUNIOR SCHOOL EDUCATION ASSESSMENT

GRADE 9 2025



INTEGRATED SCIENCE PAPER 1 (THEORY)

Time: 1 hour 30 Minutes

Name:

School:

Assessment Number:

Signature: Date:

Instructions for Learners

1. Write your name, school name, and assessment number in the spaces provided.
2. This paper consists of **29 questions**.
3. Check the question paper to ascertain all pages are printed and no questions are missing.
4. Answer all the questions in the answer booklet provided.

SCORE GRID

Question	Score	Question	Score	Question	Score
1–20		27			
21		28			
22		29			
23		30			
24					
25					
26					

ASSESSMENT RUBRICS

Learning Area	Exceeds Expectations (4)	Meets Expectations (3)	Approaching Expectations (2)	Below Expectations (1)
Integrated Science				
Learners score				

SECTION A: MULTIPLE CHOICE (20 Marks)

Answer all the questions.

1. During games time, a pile of paper and wooden furniture caught fire in a residential house near the school. Which class of fire is this and which extinguisher should be used?

A. Class A; water-based extinguisher
B. Class B; foam extinguisher
C. Class C; CO₂ extinguisher
D. Class D; dry powder extinguisher

2. David observed a chemical bottle in the laboratory with a safety symbol.



What does it represent?

A. Flammable
B. Corrosive
C. Toxic
D. Irritant

3. Sarah mixed water and alcohol in class. Which method is most suitable for separating them?

A. Chromatography
B. Simple distillation
C. Filtration
D. Sublimation

4. A container holds water up to 3 m. Calculate the pressure at the bottom (density = 1000 kg/m³, g = 9.8 m/s²).

A. 30,000 Pa B. 20,000 Pa
C. 29,400 Pa D. 39,200 Pa

5. A teacher opened a perfume bottle. After some time, the scent spread across the room. Which process is this?

A. Osmosis
B. Diffusion
C. Filtration
D. Evaporation

6. An engineer requires a metal that withstands high temperature and mechanical stress for engine parts. Which is most suitable?

A. Aluminium
B. Cast iron
C. Steel
D. Copper

7. In the food chain:

Maize → Mice → Snake → Hawk

Which animal is at the same level as the mice?

A. Goat B. Lion
C. Bacteria D. Vulture

8. Which of the following is reusing water?

A. Mulching plants
B. Using rinsing water to clean toilets
C. Turning off taps when not in use
D. Boiling dirty water and cooling the steam

9. Which statement is true about reproduction in human beings?

A. Zygote can implant anywhere in uterine wall
B. Gestation is about 12 months
C. Fertilisation occurs after implantation
D. Fertilisation occurs in the uterus

10. A grassland had chameleons, flowers, butterflies, and hawks. Which organism was most abundant?

- A. Chameleons B. Hawks
C. Flowers D. Butterflies

11. Four water samples were boiled to dryness. Which sample is likely rainwater?

Water Sample	Salt (g)
A	3.0
B	2.5
C	0.8
D	4.2

- A. C B. B C. D D. A

12. Equal soap was added to water samples X and Y.

- i. X had many minerals
ii. Y formed suspended particles.

Which statement is correct?

- A. X may have many minerals
B. Y may form lather after boiling
C. X can form scales in pipes
D. Y does not waste soap

13. Which group consists of insect-pollinated flower features only?

- (i) Sticky stigma
(ii) Thick cuticle
(iii) Large flower
(iv) Few heavy pollen
(v) No nectar

- A. (ii), (iii), (v) B. (ii), (iv), (v)
C. (i), (ii), (iv) D. (i), (iii), (iv)

14. Arrange the processes in seed formation in the correct order:

- (i) Growth of pollen tube
(ii) Pollination
(iii) Fusion
(iv) Pollen tube breaks

- A. (ii), (i), (iv), (iii)
B. (i), (ii), (iv), (iii)
C. (ii), (i), (iii), (iv)
D. (iv), (i), (iii), (ii)

15. Which is correctly matched?

- A. Needle-like leaves – reflect sunlight
B. Deep roots – reduce water loss
C. Normal hairs – trap moisture
D. Underside stomata – increase water loss

16. Which processes cause warming to the surroundings?

- A. Evaporation and freezing
B. Melting and condensation
C. Freezing and condensation
D. Melting and evaporation

17. Grade 9 learners measured force. Arrange the steps:

- (i) Suspend material on spring balance
(ii) Take reading
(iii) Tie string on material
(iv) Record results

- A. (iii), (i), (ii), (iv)
B. (iv), (iii), (i), (ii)
C. (i), (iii), (ii), (iv)
D. (ii), (iii), (iv), (i)

18. Which disease matches the signs: painless sores, sore throat, blindness, skin rashes?

- A. Chancroid B. Gonorrhoea
- C. HIV/AIDS D. Syphilis

19. Learners used nails, wood, tins, and strings to construct a simple machine. What was it used for?

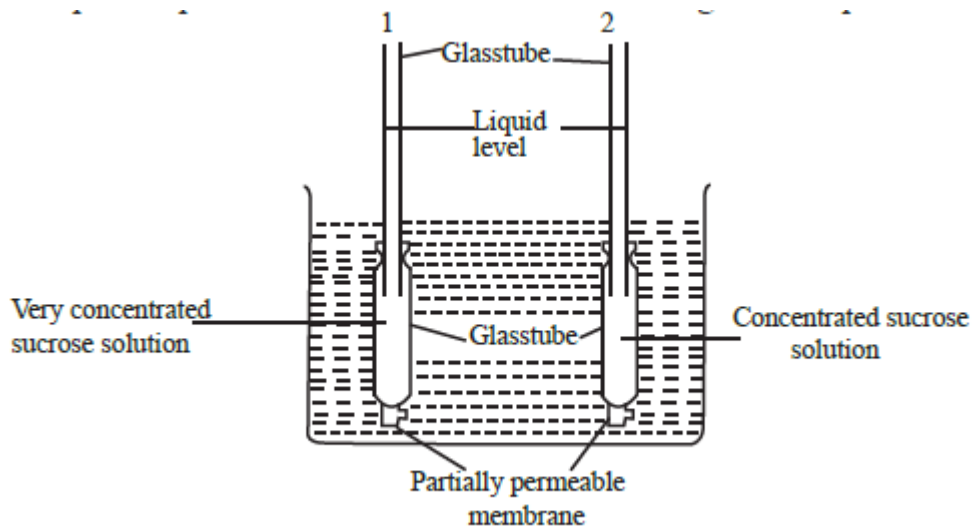
- A. Compare masses
- B. Measure force in lifting objects
- C. Balance pupils
- D. Measure wind strength

20. Which are insect-pollinated flower characteristics only?

- A. (ii), (iii), (v)
- B. (ii), (iv), (v)
- C. (i), (ii), (iv)
- D. (i), (iii), (iv)

SECTION B: STRUCTURED QUESTIONS (30 Marks)

21. Learners set up an osmosis experiment with concentrated sucrose solution separated by a partially permeable membrane.



(a) State the observation after 30 minutes. (1 mark)

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(b) Explain the observation. (2 marks)

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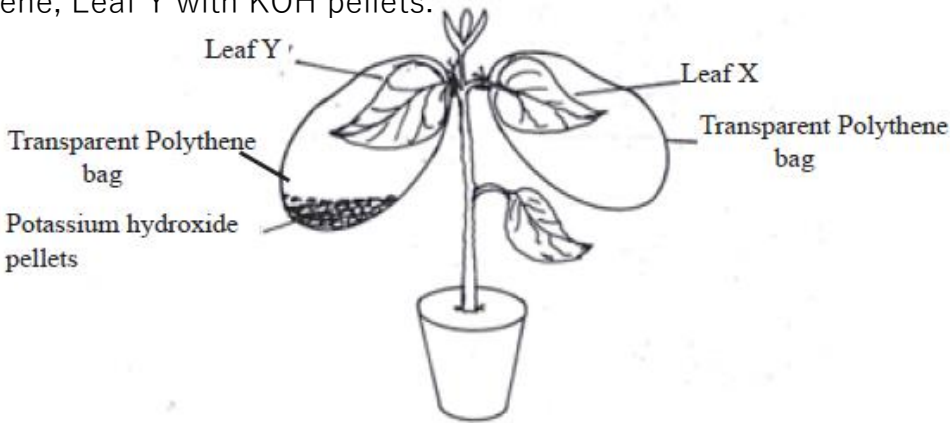
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22. Naomi studied a leaf with loosely arranged spongy mesophyll. What is the significance of this arrangement? (2 marks)

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23. A potted plant was destarched and two leaves (X and Y) treated. Leaf X was covered with polythene, Leaf Y with KOH pellets.



(a) State the starch test results for leaves X and Y. (2 marks)

.....

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(b) Give reasons for the results. (2 marks)

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24. Sodium chloride was contaminated with copper (II) oxide. Describe how pure sodium chloride can be obtained. (3 marks)

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25. A metal P reacted with steam forming oxide Q (yellow when hot, white when cool) and gas R.
(a) Identify P, Q, and R. (3 marks)

P:

Q:

R:

(b) Write the chemical equation. (1 mark)

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26. Study the properties of metals:

Metal	Density	Tensile strength	Electrical conductivity
Al	2.70	6.9	0.38
Fe	7.86	20.8	0.10
Cu	8.92	12.9	0.60

(a) Why do some stainless steels have similar properties to iron? (1 mark)

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(b) Why is aluminium with a steel core used for power cables instead of copper? (2 marks)

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(c) Why is copper used in almost all electrical uses? (1 mark)

27. The elements below have atomic numbers:

Element	X	Y	Z
Atomic number	14	17	19

(a) Which element is a poor conductor? (1 mark)

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(b) Write the electronic configuration of X. (1 mark)

.....

(c) Draw the electronic structure of Z. (2 marks)

28. Tamarah and Leonard found a cactus-like plant.



(a) Identify its habitat. (1 mark)

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(b) Give one reason. (1 mark)

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29. During cleanup, learners collected plastics, glass, food leftovers, and paper.

(a) Classify into biodegradable and non-biodegradable. (2 marks)

Biodegradable	Non-biodegradable

(b) Suggest one disposal method for biodegradable waste. (1 mark)

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(c) Why is separating waste important? (1 mark)

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ANSWERS

SECTION A: Multiple Choice Questions (20 Marks)

Qn	Correct Answer	Explanation
1	A. Class A; water-based extinguisher	Class A fires involve paper/wood; water is used.
2	C. Toxic	The bottle symbol shows toxic substance
3	B. Simple distillation	Separates miscible liquids like alcohol and water.
4	C. 29,400 Pa	$P = h \rho g = 3 \times 1000 \times 9.8 = 29,400 \text{ Pa}$.
5	B. Diffusion	Perfume spreads by random molecular movement.
6	C. Steel	Strong, withstands heat and stress in engines.
7	A. Goat	Like mice, goat is a primary consumer (herbivore).
8	B. Using rinsing water to clean toilets	That is reusing water.
9	A. Implantation	Attachment of the zygote to the walls of the uterus
10	C. Flowers	Plants are most abundant producers.
11	A. C	Rainwater has least salts (0.8 g).
12	C. X can form scales in pipes	Hard water causes scaling.
13	D. (i), (iii), (iv)	Features of insect-pollinated flowers.
14	C. (ii), (i), (iii), (iv)	Correct order of fertilisation events.
15	C. Normal hairs – trap moisture	Adaptation for water conservation.
16	C. Freezing and condensation	Both release heat to surroundings.
17	A. (iii), (i), (ii), (iv)	Correct order of measuring force.
18	D. Syphilis	Symptoms: sores, rash, blindness.
19	B. Measure force in lifting objects	Simple machine with pulleys/springs.
20	D. (i), (iii), (iv)	Insect-pollinated flowers: sticky stigma, large flowers, heavy pollen.

SECTION B: Structured Questions (30 Marks)

Q21. Osmosis Experiment

(a) Observation

- Level of solution increases on the sucrose side.

(b) Explanation

- i. Water molecules move from dilute solution to concentrated sucrose solution.
- ii. Movement is through the partially permeable membrane.
- iii. This is due to osmosis.

Q22. Spongy Mesophyll Arrangement

- ✓ Provides large air spaces for gas exchange.
- ✓ Facilitates diffusion of oxygen, carbon dioxide, and water vapour.

Q23. Photosynthesis Test

(a) Results

- Leaf X (covered with polythene): Tests positive for starch.
- Leaf Y (with KOH): Tests negative for starch.

(b) Reasons

- Leaf X received carbon dioxide and carried out photosynthesis.
- Leaf Y lacked carbon dioxide due to KOH absorption, hence no starch formed.

Q24. Separation of Sodium Chloride and Copper (II) Oxide

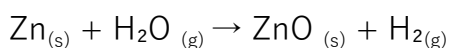
- Add water to dissolve sodium chloride (copper (II) oxide is insoluble).
- Filter to remove insoluble copper (II) oxide residue.
- Evaporate the filtrate to obtain pure sodium chloride crystals.

Q25. Metal Reaction with Steam

(a) Identification

- ✓ P: Zinc
- ✓ Q: Zinc oxide (yellow when hot, white when cool)
- ✓ R: Hydrogen gas

(b) Equation



Q26. Properties of Metals

(a) Some stainless steels have similar properties to iron because:

- They contain a large proportion of iron.

(b) Aluminium with steel core is used for power cables because:

- Aluminium is light and a good conductor.
- Steel core provides strength and prevents sagging.

(c) Copper is widely used in electrical wiring because:

- It has high electrical conductivity.
- It is ductile and easy to draw into wires.

Q27. Elements (X=14, Y=17, Z=19)

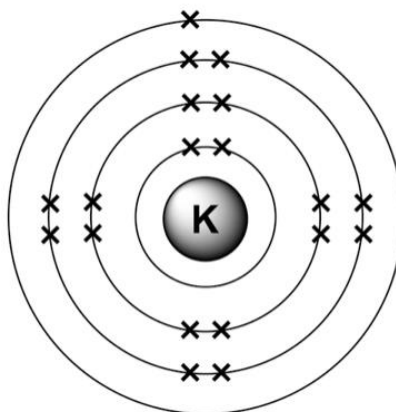
(a) Poor conductor: **Y (Chlorine)**.

(b) Electronic configuration of X (Silicon, 14):

2, 8, 4

(c) Electronic structure of Z (Potassium, 19):

- First shell: 2 electrons
- Second shell: 8 electrons
- Third shell: 8 electrons
- Fourth shell: 1 electron



Q28. Cactus-like Plant

(a) Habitat: Desert.

(b) Reason:

- Has adaptations to conserve water (thick stems, spines, deep roots).

Fig 1. Potassium. Electron shells of potassium.
Potassium has an electronic configuration of 2, 8, 8, 1.

Q29. Waste Management

(a) Classification

- **Biodegradable:** Food leftovers, paper.
- **Non-biodegradable:** Plastics, glass.

(b) Disposal method for biodegradable waste

- Composting or feeding animals.

(c) Importance of separating waste

- ✓ Facilitates recycling.
- ✓ Reduces pollution.
- ✓ Prevents spread of disease.
- ✓ Makes disposal easier.